Takaki Akiba, Ph.D. (Engineering)

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Research Interests

- Quantum computing for reacting flow problems
- Combustion limits and flame dynamics near the fuel-lean limit

Appointments

- **Apr. 2024 - Present** - Assistant Professor, Department of Mechanical Engineering, The University of Tokyo

Education

- **Mar. 2024** - Ph.D. (Engineering), Graduate School of Engineering, Tohoku University, Japan
- **Dec. 2022 - Sep. 2023** - Visiting Student Research Collaborator, Mechanical and Aerospace Engineering, Princeton University, USA
- **Mar. 2020** - M.Eng., Graduate School of Engineering, Tohoku University, Japan
- **Mar. 2018** - B.Eng., School of Engineering, Tohoku University, Japan

Fellowships

- **Apr. 2021 - Sep. 2023** - JSPS Research Fellowship for Young Scientists (DC2)
- **Apr. 2020 - Mar. 2023** - Amano Institute of Technology Scholarship

Awards

- **Aug. 2026** - Best Poster Award, 18th International Heat Transfer Conference
- **Oct. 2025** - Young Researcher Presentation Award, Thermal Engineering Division, Japan Society of Mechanical Engineers
- **Jul. 2025** - Exploratory Research Award, 2025 JHPCN Symposium
- **Mar. 2024** - Dean's Award, Graduate School of Engineering, Tohoku University
- **Mar. 2018** - Hatakeyama Award, Japan Society of Mechanical Engineers

Research Profiles

ResearchGate · GitHub · ORCID · researchmap

Publications

Peer-reviewed Journal Articles

- Xingchang Xu, Minhyeok Lee, **Takaki Akiba**, Yuji Suzuki, Quantitative measurements of hydrogen atom concentrations in ammonia diffusion flames using two-photon absorption laser-induced fluorescence, Proceedings of the Combustion Institute, 42, 2026, 106652, doi:10.1016/j.proci.2026.106652.
- Yujian Xing, Minhyeok Lee, Xingchang Xu, **Takaki Akiba**, Keigo Matsumoto, Shugo Iwasaki, Yuji Suzuki, Nitriding of Austenitic Stainless Steel Induced by Ammonia Flames under Elevated Pressures , Proceedings of the Combustion Institute, 42, 2026, 106480, doi:10.1016/j.proci.2026.106480.
- Kaito Nishiyama, Joel Daou, Akira Tsunoda, **Takaki Akiba**, Youhi Morii, Kaoru Maruta, Two-dimensional simulations of effect of stretch on ball-like flame in weakly stretched counterflow field using low-Lewis number premixture, Proceedings of the Combustion Institute, 42, 2026, 106322, doi:10.1016/j.proci.2026.106322.
- **Takaki Akiba**, Youhi Morii, Minhyeok Lee, Kaoru Maruta, Yuji Suzuki, Efficient evaluation of Arrhenius rates for quantum computing applications in reactive flow problems using Carleman linearization, Proceedings of the Combustion Institute, 41, 2025, 105918, doi:10.1016/j.proci.2025.105918.

- Andy Thawko, **Takaki Akiba**, Ziyu Wang, Bowen Mei, Wenbin Xu, Kaoru Maruta, Yiguang Ju, Low-temperature reactivity, extinction, and heat release rate of non-premixed cool flame at elevated pressures, Combustion and Flame, 270, 2024, 113727, doi:10.1016/j.combustflame.2024.113727.
- **Takaki Akiba**, Akira Tsunoda, Takuya Tezuka, Youhi Morii, Hisashi Nakamura, Kaoru Maruta, DMD analysis on sporadic flame behaviors in low-Lewis-number counterflow under microgravity, Proceedings of the Combustion Institute, 40, 2024, 105233, doi:10.1016/j.proci.2024.105233.
- **Takaki Akiba**, Youhi Morii, Kaoru Maruta, Carleman linearization approach for chemical kinetics integration toward quantum computation, Scientific Reports, 13(1), 2023, 3935, doi:10.1038/s41598-023-31009-9.
- Akira Tsunoda, **Takaki Akiba**, Hisashi Nakamura, Youhi Morii, Takuya Tezuka, Kaoru Maruta, Computational study on lean and rich combustion of flame ball, counterflow flame and planar flame: Their limits and stoichiometries, Proceedings of the Combustion Institute, 39(2), 2022, 1937-1944, doi:10.1016/j.proci.2022.07.158.
- **Takaki Akiba**, Tomoya Okuno, Hisashi Nakamura, Youhi Morii, Takuya Tezuka, Roman Fursenko, Sergey Minaev, Masao Kikuchi, Kaoru Maruta, Dynamics of ball-like flames in extremely low-speed counterflow field in near-lean limit low-Lewis number mixture, Proceedings of the Combustion Institute, 38(2), 2020, 1965-1972, doi:10.1016/j.proci.2020.06.084.
- Tomoya Okuno, **Takaki Akiba**, Hisashi Nakamura, Roman Fursenko, Sergey Minaev, Takuya Tezuka, Susumu Hasegawa, Masao Kikuchi, Kaoru Maruta, Broken C-shaped extinction curve and near-limit flame behaviors of low Lewis number counterflow flames under microgravity, Combustion and Flame, 194, 2018, 343-351, doi:10.1016/j.combustflame.2018.05.014.

Preprints

- **Takaki Akiba**, Youhi Morii, Duplicate-Aware Shift-and-Lift Carleman Linearization: Structure, Complexity, and Comparative Evaluation, arXiv, 2026, doi:10.48550/arXiv.2605.05901.
- **Takaki Akiba**, Youhi Morii, GPU Benchmark through QPE Emulator with cuQuantum for Practical Quantum Applications, arXiv, 2025, doi:10.48550/arXiv.2507.17175.
- **Takaki Akiba**, Youhi Morii, Kaoru Maruta, Carleman linearization approach for chemical kinetics integration toward quantum computation, arXiv, 2022, doi:10.48550/arXiv.2207.01818.

Conference Presentations

International Conference Presentations

- **Takaki Akiba**, Youhi Morii, GPU-Based Implementations of QPE and HHL with cuQuantum for Practical Quantum Emulation, IEEE Quantum Week 2026, QAPP-1207, September 13-18, 2026, Toronto, Canada (Oral).
- **Takaki Akiba**, Youhi Morii, Minhyeok Lee, Kaoru Maruta, Yuji Suzuki, Quantum algorithm with Carleman-Taylor linearization for reacting flow, 18th International Heat Transfer Conference, IHTC18-0836, August 2-7, 2026, Rio de Janeiro, Brazil (Poster).
- Shusuke Umeki, Gyeonghun Yoo, **Takaki Akiba**, Minhyeok Lee, Yuji Suzuki, TDLAS measurements of flame temperature and CO concentration for off-gas combustion of SOFC, 41st International Symposium on Combustion, F-06-10, July 26-31, 2026, Kyoto, Japan (Poster).
- Aki Fujinawa, **Takaki Akiba**, Minhyeok Lee, Yuji Suzuki, Ewa Marek, On the combustion of hydrogen in a fluidised bed of silica revealed by OH* chemiluminescence and high-speed imaging, 41st International Symposium on Combustion, 5-B2-09, July 26-31, 2026, Kyoto, Japan (Oral).
- Yujian Xing, Minhyeok Lee, **Takaki Akiba**, Keigo Matsumoto, Shugo Iwasaki, Yuji Suzuki, Nitriding of Austenitic Stainless Steel Induced by Ammonia Flames under Elevated Pressures, 41st International Symposium on Combustion, 5-C1-01, July 26-31, 2026, Kyoto, Japan (Oral).
- Xingchang Xu, Minhyeok Lee, **Takaki Akiba**, Yuji Suzuki, Quantitative measurements of hydrogen atom concentrations in ammonia diffusion flames using two-photon absorption laser-induced fluorescence, 41st International Symposium on Combustion, 2-C1-01, July 26-31, 2026, Kyoto, Japan (Oral).
- Rupam Islam, Xingchang Xu, **Takaki Akiba**, Minhyeok Lee, Yuji Suzuki, Quantitative NH_2 radical measurements in ammonia flames using planar laser-induced fluorescence, 41st International Symposium on Combustion, M-06-06, July 26-31, 2026, Kyoto, Japan (Poster).
- **Takaki Akiba**, Youhi Morii, Minhyeok Lee, Kaoru Maruta, Yuji Suzuki, Linearization method via Carleman-Taylor coupling toward quantum computing of reacting flow, APS Division of Fluid Dynamics Annual Meeting 2025, November 23-25, 2025, Houston, Texas (Oral).
- **Takaki Akiba**, Youhi Morii, Minhyeok Lee, Kaoru Maruta, Yuji Suzuki, Carleman Linearization of Reacting Flow Problems toward Quantum Computing, 22nd International Conference on Flow Dynamics, OS2-9, November 10-14, 2025, Sendai, Japan (Oral).
- Xingchang Xu, Minhyeok Lee, **Takaki Akiba**, Yuji Suzuki, Quantitative Measurements of H-atom and OH in Ammonia Counterflow Diffusion Flames, 4th Symposium on Ammonia Energy, September 28-October 2, 2025, Minneapolis, Minnesota (Oral).

- **Takaki Akiba**, Youhi Morii, Minhyeok Lee, Kaoru Maruta, Yuji Suzuki, Carleman Linearization Application for Chemical Reaction and Reacting Flow toward Quantum Computing, IEEE Quantum Week 2025, 13-1205, August 31-September 5, 2025, Albuquerque, New Mexico (Poster).
- **Takaki Akiba**, Youhi Morii, Minhyeok Lee, Kaoru Maruta, Yuji Suzuki, Efficient evaluation of Arrhenius rates for quantum computing applications in reactive flow problems using Carleman linearization, 15th Asia-Pacific Conference on Combustion, 372, May 19-22, 2025, Singapore (Oral).
- Xingchang Xu, Minhyeok Lee, **Takaki Akiba**, Yuji Suzuki, Quantitative measurements of hydrogen atom concentrations in ammonia diffusion flames using two-photon absorption laser-induced fluorescence, 15th Asia-Pacific Conference on Combustion, 460, May 19-22, 2025, Singapore (Oral).
- Hyunkyu Park, **Takaki Akiba**, Minhyeok Lee, Yuji Suzuki, OH Tunable Diode Laser Absorption Spectroscopy for Temperature Measurements in Hydrogen Flames, 62nd NHTS/HTSJ International Heat Transfer Symposium, IOS7-2-05, May 7-10, 2025, Okinawa (Oral).
- **Takaki Akiba**, Youhi Morii, Minhyeok Lee, Kaoru Maruta, Yuji Suzuki, Toward quantum computing of nonlinear reacting flows with Carleman linearization, The 3rd Pacific Rim Thermal Engineering Conference, PRTEC-241215, December 15-19, 2024, Hawaii, USA (Oral).
- Xingchang Xu, **Takaki Akiba**, Minhyeok Lee, Yuji Suzuki, Investigation of ammonia counterflow diffusion flames through direct measurements of species distribution, 40th International Symposium on Combustion, WiPP-2P095, July 21-27, 2024, Milan, Italy (Poster).
- **Takaki Akiba**, Akira Tsunoda, Takuya Tezuka, Youhi Morii, Hisashi Nakamura, Kaoru Maruta, DMD analysis on sporadic flame behaviors in low-Lewis-number counterflow under microgravity, 40th International Symposium on Combustion, 2C04, July 21-26, 2024, Milan, Italy (Oral).
- **Takaki Akiba**, Akira Tsunoda, Youhi Morii, Hisashi Nakamura, Kaoru Maruta, DMD analysis on quasi-steady behaviors of sporadic flames in counterflow under low-speed low-Lewis-number conditions, 19th International Conference on Numerical Combustion, ICNC2024-1097, May 7-10, 2024, Kyoto, Japan (Oral).
- Andy Thawko, Ziyu Wang, **Takaki Akiba**, Bowen Mei, Yiguang Ju, Study on the low-temperature chemistry and Heat Release Rate of High-pressure Diethyl Ether diffusion Cool Flame, AIAA SCITECH 2024 Forum, January 8-12, 2024, Orlando, Florida (Oral).
- Akira Tsunoda, **Takaki Akiba**, Hisashi Nakamura, Takuya Tezuka, Masao Kikuchi, Kaoru Maruta, L3-FLAME: Low-speed Low-Lewis-number Counterflow Flame Experiment for Unified Combustion Limit Theory - Preliminary Airplane-based Microgravity Experiments and Computational Study-, 34th International Symposium on Space Technology and Science & 12th Nano-Satellite Symposium(ISTS2023), 2023-h-06, June 3-9, 2023, Kurume, Japan (Oral).
- Akira Tsunoda, **Takaki Akiba**, Hisashi Nakamura, Youhi Morii, Takuya Tezuka, Kaoru Maruta, Near Lean/Rich Limits Behaviors and Local Stoichiometries of Flame Balls, Counterflow Flames, and Planar Flames in a CH4/O2/Xe Mixture, 19th International Conference on Flow Dynamics, OS2-21, November 11, 2022, Sendai, Japan (Oral).
- Kazutaka Sagawa, **Takaki Akiba**, Akira Tsunoda, Youhi Morii, Hisashi Nakamura, Kaoru Maruta, Numerical Analysis of Flame Behavior Initiated from Flame Ball and Hot Spherical Zone in Counterflow Field, 19th International Conference on Flow Dynamics, OS2-20, November 9-11, 2022, Sendai, Japan (Oral).
- **Takaki Akiba**, Youhi Morii, Kaoru Maruta, Numerical Integration Approach with Carleman Linearization for Chemical Reactions and Combustion toward Quantum Computation, 19th International Conference on Flow Dynamics, OS2-3, November 9-11, 2022, Sendai, Japan (Oral).
- Kazutaka Sagawa, **Takaki Akiba**, Takuya Tezuka, Youhi Morii, Hisashi Nakamura, Kaoru Maruta, Numerical analysis of flame behavior initiated from flame ball in weakly-stretched counterflow field, 39th International Symposium on Combustion, 2P114, July 25-29, 2022, Vancouver, Canada (Poster).
- **Takaki Akiba**, Youhi Morii, Kaoru Maruta, Numerical method for ODEs of chemical reactions with Carleman linearization toward quantum computings, 39th International Symposium on Combustion, 1P002, July 25-29, 2022, Vancouver, Canada (Poster).
- Akira Tsunoda, **Takaki Akiba**, Hisashi Nakamura, Youhi Morii, Takuya Tezuka, Kaoru Maruta, Computational study on lean and rich combustion of flame ball, counterflow flame and planar flames: Their limits and stoichometries, 39th International Symposium on Combustion, 1B10, July 25-29, 2022, Vancouver, Canada (Oral).
- **Takaki Akiba**, Youhi Morii, Kaoru Maruta, Numerical method using Carleman linearization for ODEs of chemical reactions toward quantum computation, 18th International Conference on Numerical Combustion, 3C17, May 8-11, 2022, San Diego, California (Oral).
- Akira Tsunoda, **Takaki Akiba**, Hisashi Nakamura, Takuya Tezuka, Kaoru Maruta, Computational Study on Flame Balls at Fuel Lean and Rich Conditions, 18th International Conference on Flow Dynamics, OS2-17, October 27-29, 2021, Sendai, Japan (Oral).
- **Takaki Akiba**, Akira Tsunoda, Hisashi Nakamura, Takuya Tezuka, Masao Kikuchi, Kaoru Maruta, Counterflow premixed flame experiments at ISS Kibo for comprehensive combustion limit theory, 18th International Conference on Flow Dynamics, OS2-18, October 27-29, 2021, Sendai, Japan (Oral).

- Kazutaka Sagawa, **Takaki Akiba**, Youhi Morii, Hisashi Nakamura, Kaoru Maruta, Numerical Analysis of Flame Behavior Initiated from Flame Ball in Counterflow Field, 18th International Conference on Flow Dynamics, OS2-7, October 27-29, 2021, Sendai, Japan (Oral).
- **Takaki Akiba**, Tomoya Okuno, Hisashi Nakamura, Youhi Morii, Takuya Tezuka, Roman Fursenko, Sergey Minaev, Masao Kikuchi, Kaoru Maruta, Dynamics of ball-like flames in extremely low-speed counterflow field in near-lean limit low-Lewis number mixture, 38th International Symposium on Combustion, 2B14, January 25-29, 2021, Adelaide, Australia (online) (Oral).
- Akira Tsunoda, **Takaki Akiba**, Hisashi Nakamura, Takuya Tezuka, Kaoru Maruta, Computational flame regime of flame ball, counterflow and planar flames in fuel lean/ rich mixtures at low Lewis numbers, 38th International Symposium on Combustion, LF-01, January 25-29, 2021, Adelaide, Australia (online) (Poster).
- **Takaki Akiba**, Tomoya Okuno, Hisashi Nakamura, Youhi Morii, Takuya Tezuka, Roman Fursenko, Sergey Minaev, Masao Kikuchi, Kaoru Maruta, Mechanism of low-Lewis-number and near-limit ball-like flame splitting in low-speed counterflow field, 17th International Conference on Flow Dynamics, OS2-13, October 28-30, 2020, Sendai, Japan (online) (Oral).
- **Takaki Akiba**, Tomoya Okuno, Hisashi Nakamura, Youhi Morii, Takuya Tezuka, Roman Fursenko, Sergey Minaev, Masao Kikuchi, Kaoru Maruta, Dynamic behaviors of flame ball in flow, 16th International Conference on Flow Dynamics, OS2-20, November 6-8, 2019, Sendai, Japan (Oral).
- **Takaki Akiba**, Tomoya Okuno, Hisashi Nakamura, Takuya Tezuka, Susumu Hasegawa, Roman Fursenko, Sergey Minaev, Masao Kikuchi, Kaoru Maruta, Effect of diffusive thermal instability on intrinsic limits of counterflow flames, unstrained planar flames and flame ball, 12th Asia-Pacific Conference on Combustion, 1224, July 1-5, 2019, Fukuoka, Japan (Oral).
- **Takaki Akiba**, Tomoya Okuno, Hisashi Nakamura, Takuya Tezuka, Susumu Hasegawa, Roman Fursenko, Sergey Minaev, Masao Kikuchi, Kaoru Maruta, 3D computations of combustion limit phenomena of low Lewis number mixture under gravity-free condition,, 17th International Conference on Numerical Combustion, May 6-8, 2019, Aachen, Germany (Oral).
- **Takaki Akiba**, Tomoya Okuno, Susumu Hasegawa, Hisashi Nakamura, Roman Fursenko, Sergey Minaev, Masao Kikuchi, Kaoru Maruta, Destabilization of weakly stretched counterflow flames at low Lewis number under microgravity, 15th International Conference on Flow Dynamics, OS2-77, November 7-9, 2018, Sendai, Japan (Oral).
- **Takaki Akiba**, Tomoya Okuno, Hisashi Nakamura, Takuya Tezuka, Susumu Hasegawa, Masao Kikuchi, Kaoru Maruta, Near-limit behaviors of low-Lewis-number premixed counterflow flames under microgravity, 37th International Symposium on Combustion, 2P097, July 29-August 3, 2018, Dublin, Ireland (Poster).

Last updated: 2026-10-13